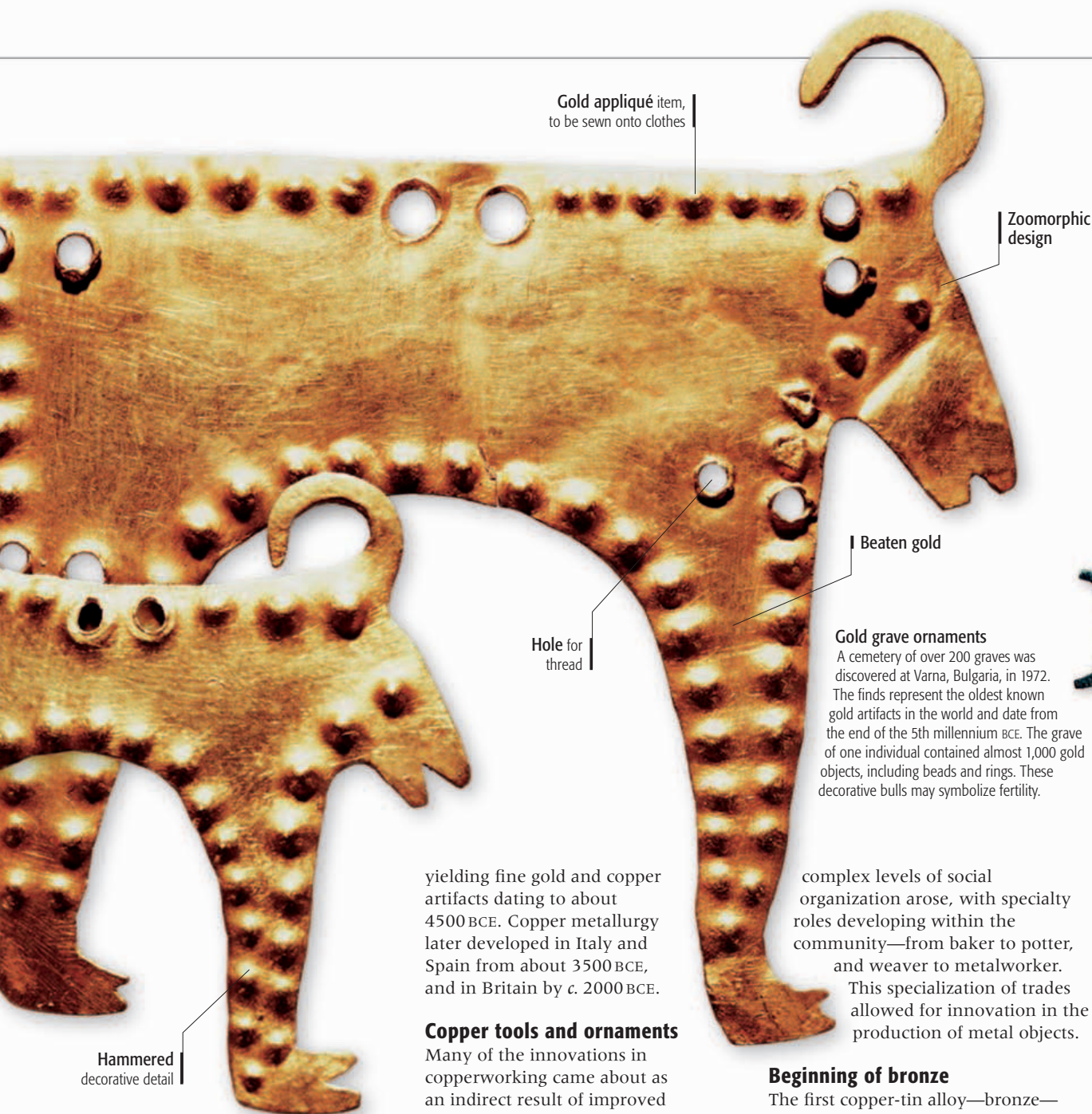




Gold appliqué item,
to be sewn onto clothes

Zoomorphic
design

Beaten gold

Hole for
thread

Hammered
decorative detail

yielding fine gold and copper artifacts dating to about 4500 BCE. Copper metallurgy later developed in Italy and Spain from about 3500 BCE, and in Britain by c. 2000 BCE.

Copper tools and ornaments

Many of the innovations in copperworking came about as an indirect result of improved agricultural methods. Efficient arable farming in the Nile valley enabled villages to develop and thrive, maintained by a surplus of food. More efficient farming methods meant that not everyone was needed for food production, and so more

complex levels of social organization arose, with specialty roles developing within the community—from baker to potter, and weaver to metalworker. This specialization of trades allowed for innovation in the production of metal objects.

Beginning of bronze

The first copper-tin alloy—bronze—was produced in around 2500 BCE, although it did not come into widespread use until much later. Bronze is harder than copper and therefore more suitable for practical applications, such as weapons, armor, and tools.

The use of metal, from bronze to iron and steel, has transformed human civilization.

BEAKER CULTURE

Between 2800 and 1900 BCE, at the beginning of the Bronze Age, people of the Beaker Culture (so named for the distinctive shape of the pottery) lived across a large area of northern Europe. Both **flint and bronze tools**, such as these daggers, are associated with this culture.



BEAKER CULTURE
TOOLS

COPPER TRADE

As one of the raw materials for bronze, copper needed to be transported in increasingly large quantities. Handles



COPPER INGOT

on each end for carrying, and a uniform size, made ingots of the metal easier to transport, and trade and quantities easier to control. This copper ingot found in a hoard from Cyprus dates from about 1200 BCE.

BRONZE AGE A period defined by the use of bronze as the most important material for making tools and weapons. In the Middle East it is approximately 3000–1200 BCE.

IRON AGE

Iron was introduced in around 1200 BCE in the Middle East and eventually became used more than bronze, as it is tougher and **more suitable for use in tools and weapons**.

EARLY STEEL

Iron can be smelted only at about 2,900°F (1,600°C), though the metal can be worked at lower temperatures. Steel, an alloy (mixture) of iron and carbon, came into limited use as early as 500 BCE in China.

GOLD FEVER

In January of 1848 a work crew camped on the American River at Coloma near Sacramento, CA found a few tiny nuggets of gold. This discovery set off a mass of migration as half a million people from around the world descended on California in search of instant wealth.



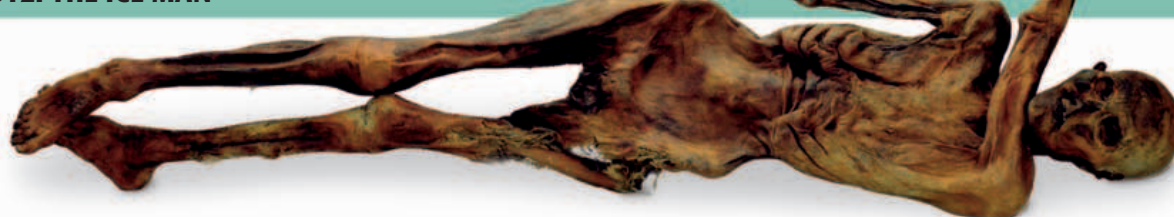
GOLD NUGGET

INDUSTRIAL REVOLUTION

Steel only became a widely **mass-produced metal** with the advent of the Bessemer converter, which was developed in 1855 during the Industrial Revolution **295 >>**

HOW WE KNOW

ÖTZI THE ICE MAN



In 1991, the frozen body of a man dating to around 3350 BCE was found preserved in a glacier. He was nicknamed Ötzi, after the Ötztal Valley of the Alps where he was found. Scientific analysis has revealed many details about his life and violent death.

Arsenic, found in his hair, showed that he had worked with copper. A copper ax, an unfinished bow and arrows, and a flint knife were found alongside his body. His last meal, containing the primitive wheat einkorn, was preserved in his intestines. At the time of death,

Ötzi was fleeing attackers—a flint arrowhead lies in one shoulder and he parried a dagger attack with his hands before dying. Ötzi's bones show he suffered from arthritis, and the tattoos found on his body may have been put there for therapeutic purposes.